



151300110071



中国认可
国际互认
检测
TESTING
CNAS L0098



TEST REI

PAGE 1 OF 32

Applicant	
Name:	Fujian GIBO Kitchen & Bath tech
Address:	Bld.54, Part B,Pushang Industrial
Post Code:	350008
Phone / Fax:	18959111663
E-mail:	lidalianq@qibol.com.cn

检 验 报 告

Test Report



福建省产品质量检验研究院

Fujian Inspection and Research Institute for Product Quality (FQII)

地址(Add): 福州市杨桥西路山头角121号 (121 Shantoujiao, West Yang Qiao Road, Fuzhou, P. R. of China)

网址(Website): www.fcii.net, www.fqii.cn

电子信箱(E-mail): quality@fcii.net



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 1 OF 32

NO. (2020)DZI0022

Applicant			
Name:	Fujian GIBO Kitchen & Bath tech Co.,Ltd		
Address:	Bld.54, Part B,Pushang Industrial Park, Fuzhou, Fujian, China		
Post Code:	350008		
Phone / Fax:	18959111663		
E-mail:	lidaliang@gibol.com.cn		
Sample(s)			
Name:	Automatic Faucet		
Type/Model:	GBL-6102AD、GBL-6110AD、GBL-6111AD、GBL-6113AD、GBL-6133AD、GBL-6150AD、GBL-6153AD、GBL-6117AD、GBL-9105AD、GBL-9160AD、GBL-9170AD		
Trademark:	GIBO		
Manufacturer:	Fujian GIBO Kitchen & Bath tech Co.,Ltd		
Amount:	1 pc		
Sample grade:	/		
Serial number:	/		
Production date:	/		
Date Received:	2020-05-08		
State Received:	The sample has well-tended appearance before the test,and work normally when it is powered on.		
Test Specification	EN61000-6-1:2007 Electromagnetic compatibility (EMC) - Part 6-1: Generic standards - Immunity for residential, commercial and light-industrial environments EN61000-6-3:2007+A1:2011 Electromagnetic compatibility (EMC) - Part 6-3: Emission - standard for residential, commercial and light-industrial environments		
Test Results	see following pages		
Conclusion	This submitted sample was deemed to have complied with the above requirements.		
Test Date(s)	2020-05-26 to 2020-05-30	Test Location	Mawei Lab
Note			

This report consists of the cover page, the title page and the body, which shall not be reproduced in part. The reproduced report is not valid unless re-affixed the "special seal for test report".The test results represented in this report relate only to the object tested, which shall not be used for improper advertising purposes without the approval of the issuing testing laboratory.

Approved by:
(signature)

张国强

Checked by:
(signature)

陈成创

Tested by:
(signature)

陈成创



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 2 OF 32

NO. (2020)DZI0022

Contents

1 General test information.....	3
4.2 General Description of Part.....	3
4.2 Description of operating modes	3
4.2 EUT operating conditions	3
4.2 Date of Order.....	3
4.2 Submitted Sample(s).....	3
4.2 Test Duration	3
4.2 Support equipment.....	3
2 Test standards and test results summary	4
4.2 Test standards	4
4.2 Test results summary	4
3 Emissions.....	6
4.2 Radiated emissions on enclosure port.....	6
4.2 Conducted emissions on low voltage AC mains port.....	9
4 Immunity.....	12
4.2 Standard compliance/severity level/criteria.....	12
4.3 Performance criterion	14
4.3 Electrostatic discharges	15
4.2 Radio-frequency electromagnetic field.....	17
4.3 Radio-frequency common mode	19
4.4 Electrical fast transient	21
4.5 Surges	23
4.6 Voltage dips and interruptions.....	25
Appendix A Equipments used during test.....	27
Appendix B EUT test photo	28
Appendix C Photo of EUT	32



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 3 OF 32

NO. (2020)DZ10022

1 General test information

4.2 General Description of Part

Product: Automatic Faucet
Manufacturer: Fujian GIBO Kitchen & Bath tech Co.,Ltd
Brand Name: GIBO
Model Number: GBL-6110AD
Rating: AC 220V 50Hz
Additional Model Number(s): GBL-6102AD、GBL-6111AD、GBL-6113AD、GBL-6133AD、GBL-6150AD、GBL-6153AD、GBL-6117AD、GBL-9105AD、GBL-9160AD、GBL-9170AD
Model Difference: Differences between the models are the name, the appearance and the decoration, do not affect the performance of EMC.
Remark:

4.2 Description of operating modes

Operating mode	Detail description
Mode 1	The normal working condition.

4.2 EUT operating conditions

The EUT exercise program used during radiated measurement was designed to exercise the various system components in a manner similar to a typical use. The standard test output as following:

I/O port type	quantity	Tested with
/	/	/

4.2 Date of Order

2020-07-07

4.2 Submitted Sample(s)

1 Sample

4.2 Test Duration

2020-05-26 to 2020-05-30

4.2 Support equipment

/



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 4 OF 32

NO. (2020)DZI0022

2 Test standards and test results summary

4.2 Test standards

EN 61000-6-1: 2007	Electromagnetic compatibility (EMC) - Part 6-1: Generic standards - Immunity for residential, commercial and light-industrial environments
EN 61000-6-3: 2007+A1:2011	Electromagnetic compatibility (EMC) - Part 6-3: Emission – standard for residential, commercial and light-industrial environments

4.2 Test results summary

Emission						
Test descriptions		Test requirements	Limit	Test results	U (dB) k=2	Remark
Radiated emissions on enclosure port	Frequency range 30MHz to 1000MHz	EN 61000-6-3: 2007+A1:2011	Class B	Pass	4.7	NOTE(4)
	Frequency range above 1000MHz		/	N/A	4.7	NOTE(15)
Conducted emissions on low voltage AC mains port	0kHz-2kHz		Class A	N/A	3.0	NOTE(2) NOTE(3) NOTE(18) NOTE(19)
	0.15MHz-30MHz		Class B	Pass	3.0	NOTE(4) NOTE(5)
Conducted emissions on DC power port			/	N/A	3.0	NOTE(4)
Conducted emissions on Telecommunications/network port			/	N/A	/	NOTE(4) NOTE(5) NOTE(6)
Immunity						
Test descriptions		Test requirements	Performance Criterion	Test results	U (dB) k=2	Remark
Electrostatic discharges		EN 61000-6-1: 2007	B	Pass	5%	
Radio-frequency electromagnetic field			A	Pass	1.9	NOTE(9)
Radio-frequency common mode			A	Pass	1.9	NOTE(10) NOTE(11)
Power-frequency magnetic field			A	N/A	N/A	NOTE(7) NOTE(8) NOTE(16)
Fast transients			B	Pass	10%	NOTE(11) NOTE(13)
Surges			B	Pass	10%	NOTE(12)
Voltage dips and interruptions			B/B/C	Pass	2%	NOTE(14)



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 5 OF 32

NO. (2020)DZ10022

Remarks:

- (1) If the internal emission source(s) is operating at a frequency below 9kHz the measurements need only to be performed up to 230MHz.
- (2) Applicable to apparatus covered within the scope of IEC 61000-3-2, IEC 61000-3-3 or IEC 61000-3-12.
- (3) If the apparatus is covered by the scope of IEC 61000-3-11, this may be used instead of IEC 61000-3-3.
- (4) At transitional frequencies the lower limit applies.
- (5) The limits decrease linearly with the logarithm of the frequency in the range 0.15MHz to 0.5MHz.
- (6) The current and voltage disturbance limits are derived for use with an impedance stabilization network (ISN) which presents a common mode (asymmetric mode) impedance of 150Ω to the telecommunication port under test.
- (7) Applicable only to apparatus containing devices susceptible to magnetic fields.
- (8) For CRTs, the acceptable jitter depends upon the character size and is calculated for a test level of 1A/m.
- (9) The frequency range has been selected to cover the frequencies with the highest potential risk of a disturbance.
- (10) The test level can also be defined as the equivalent current into a 150Ω load.
- (11) Applicable only to ports interfacing with cables whose total length according to the manufacturer's functional specification may exceed 3m.
- (12) Not applicable to input ports intended for connection to a battery or a rechargeable battery which must be removed or disconnected from the apparatus for recharging. Apparatus with a DC power input port intended for use with an AC-DC power adaptor shall be tested on the AC power input of the AC-DC power adaptor specified by the manufacturer or, where none is so specified, using a typical AC-DC power adaptor, DC ports which are not intended to be connected to a DC distribution network are treated as signal ports.
- (13) Not applicable to input ports intended for connection to a battery or a rechargeable battery which must be removed or disconnected from the apparatus for recharging. Apparatus with a DC power input port intended for use with an AC-DC power adaptor shall be tested on the AC power input of the AC-DC power adaptor specified by the manufacturer or, where none is so specified, using a typical AC-DC power adaptor, The test is applicable to DC power input ports intended to be connected permanently to cables longer than 3m.
- (14) Applicable only to input ports.
- (15) The highest internal frequency of the EUT is 8 MHz, the measurement shall only be made up to 1 GHz.
- (16) Infrared induction without Hall sensor.
- (17) N/A-Not Applicable).
- (18) The rated power of the prototype is less than 75W, the harmonic current test is not applicable to the product of this application.
- (19) Through the inspection of product manual and short-term function test, it is determined that the product is unlikely to produce serious voltage fluctuation. The voltage fluctuation and flicker test is not applicable to the product of this application.



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 6 OF 32

NO. (2020)DZI0022

3 Emissions

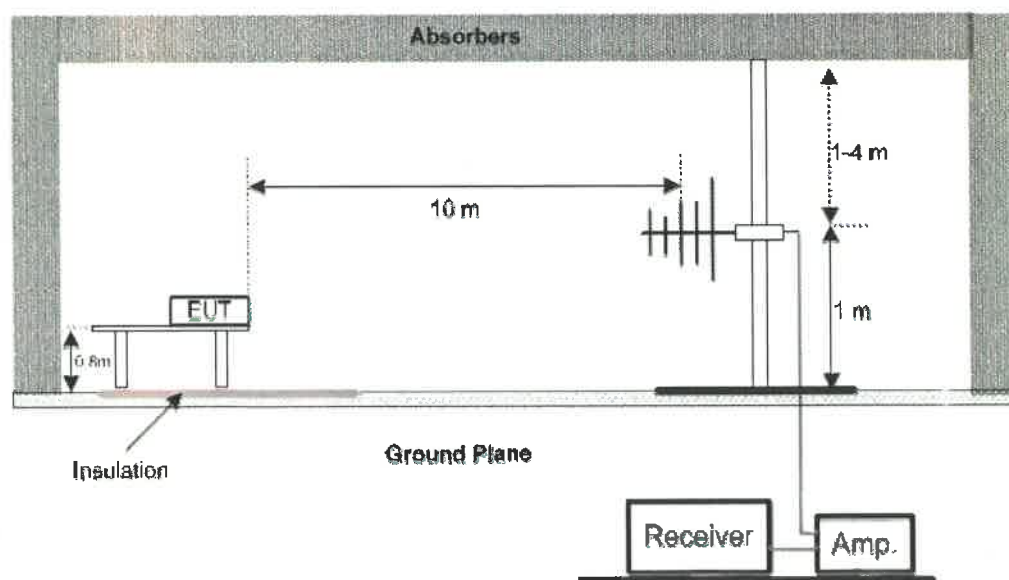
4.2 Radiated emissions on enclosure port

Standard:	EN 61000-6-3
Classification of EUT:	Class B
Test result:	Pass
Temperature:	(17~23)°C
Relative humidity:	(50~60)%
Test date:	2020-05-25
Test voltage:	AC 220V, 50Hz
Operating mode:	Mode 1

3.2.1. Test method

The test was performed in accordance with EN 61000-6-3. The prescans were performed by a peak measuring receiver. The high values found were measured by a quasi-peak measuring receiver.

3.2.1. Test setup





Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 7 OF 32

NO. (2020)DZI0022

3.2.1. Limits Requirements for radiated emissions on enclosure port at frequencies up to 1 GHz

Table clause	Frequency range MHz	Measurement			Class A limits dB(μV/m)
		Facility (see Table A.1)	Distance m	Detector type/ bandwidth	
Table 1	30 to 230	OATS/SAC	10	Quasi Peak/ 120kHz	30
	230 to 1000				37
	30 to 230	OATS/SAC	3		40
	230 to 1000				47
At transitional frequencies the lower limit applies.					



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

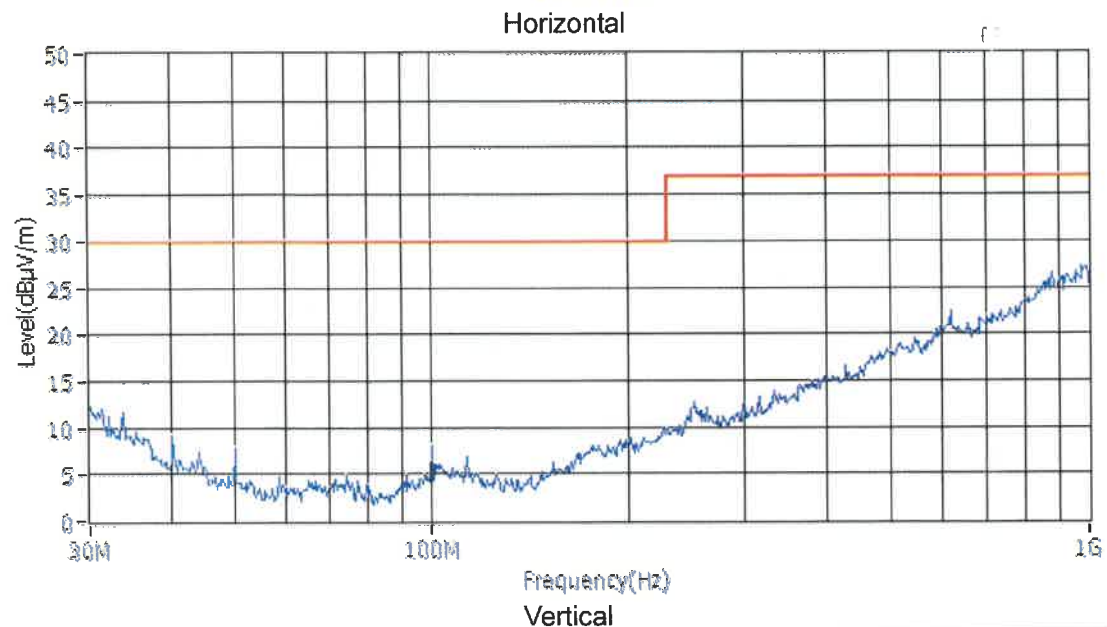
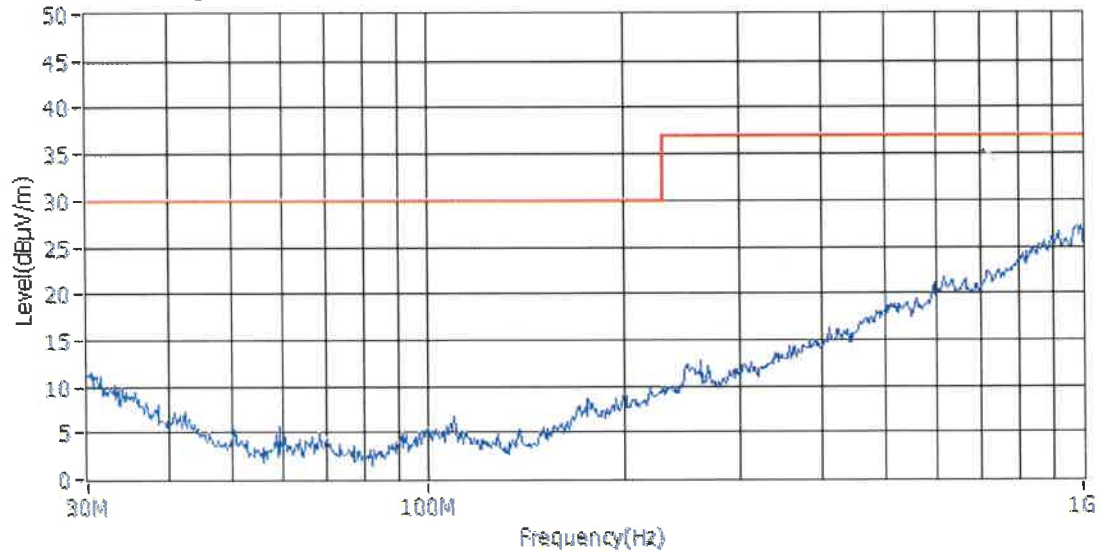
PAGE 8 OF 32

NO. (2020)DZI0022

3.2.1. Test results at 10m of SAC

Model Name	GBL-6110AD	Test Voltage	AC 220V, 50Hz
Frequency MHz	Polarity	Limit dB(μ V/m)	Measured level dB(μ V/m)
30.0000	Vertical	30	10.4
33.7000	Vertical	30	9.9
112.8500	Vertical	30	5.1
251.1500	Vertical	37	10.8
261.4000	Horizontal	37	10.9
617.8500	Vertical	37	20.5

The reference diagrams as follows: (peak)





Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 9 OF 32

NO. (2020)DZI0022

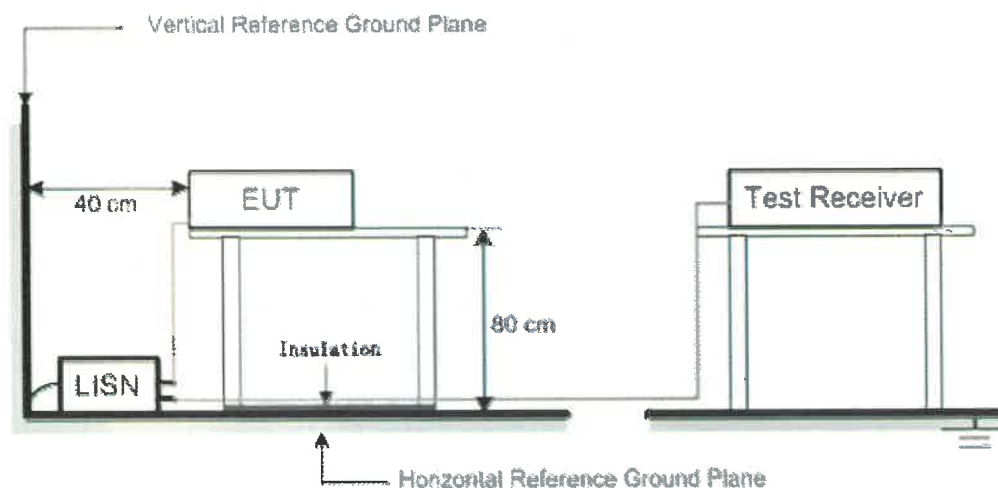
4.2 Conducted emissions on low voltage AC mains port

Standard:	EN 61000-6-3
Classification of EUT:	Class A
Test result:	Pass
Temperature:	(17~23)°C
Relative humidity:	(50~60)%
Test date:	2020-05-26
Test voltage:	AC 220V, 50Hz
Operating mode:	Mode 1

3.2.1. Test method

The test was performed in accordance with EN 61000-6-3. The prescans were performed in peak and average detection modes on each mains lead. Any emissions recorded within 20dB of the relevant limit lines were final measured using quasi-peak and average detection on all the mains leads with the worst case recorded in the table of results.

3.2.2. Test setup





Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 10 OF 32

NO. (2020)DZI0022

3.2.3 Limits

Requirements for conducted emissions from low voltage AC mains ports of class A equipment

Table clause	Frequency range MHz	Coupling device	Detector type/ bandwidth	Class A limits dB(μV)
A9.1	0.15 to 0.5	AMN	Quasi Peak/ 9kHz	79
	0.5 to 30			73
A9.1	0.15 to 0.5	AMN	Average/ 9kHz	66
	0.5 to 30			60

Apply A9.1 and A9.2 across the entire frequency range.

Requirements for conducted emissions from low voltage AC mains ports of class B equipment

Table clause	Frequency range MHz	Coupling device	Detector type/ bandwidth	Class B limits dB(μV)
A10.1	0.15 to 0.5	AMN	Quasi Peak/ 9kHz	66 to 56
	0.5 to 5			56
	5 to 30			60
A10.1	0.15 to 0.5	AMN	Average/ 9kHz	56 to 46
	0.5 to 5			46
	5 to 30			50

Apply A10.1 and A10.2 across the entire frequency range.



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 11 OF 32

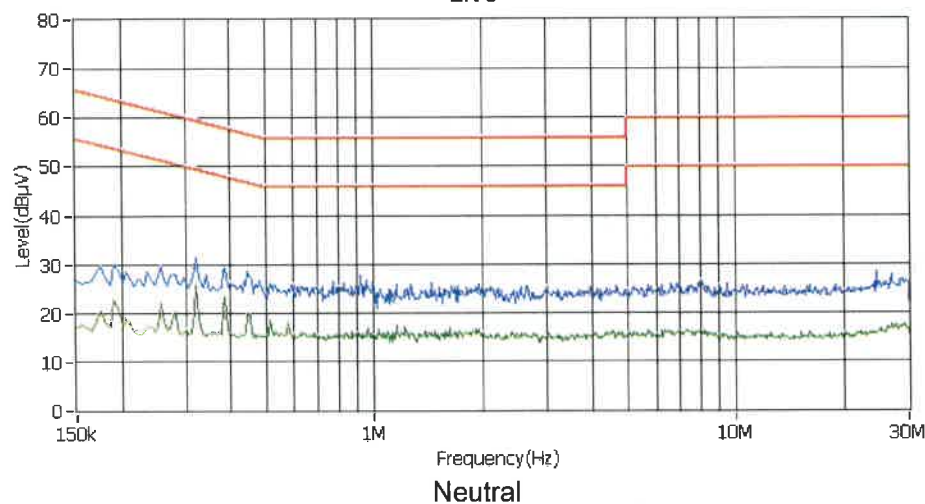
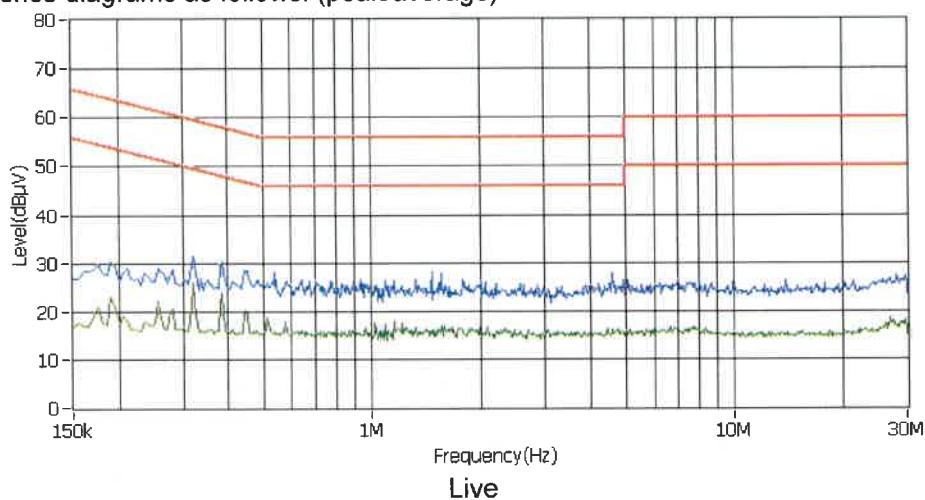
NO. (2020)DZI0022

3.2.4 Test Result

Model Name	GBL-6110AD	Test Voltage	AC 220V, 50Hz
------------	------------	--------------	---------------

Quasi-peak				Average			
Frequency MHz	Limit dB(μV)	Measured level dB(μV)	Conductor L or N	Frequency MHz	Limit dB(μV)	Measured level dB(μV)	Conductor L or N
0.255	61.6	25.4	N	0.190	54.0	23.4	L
0.320	59.7	28.1	L	0.255	51.6	22.5	N
0.380	58.3	24.7	L	0.320	49.7	26.2	L
0.445	57.0	23.1	N	0.385	48.2	23.8	L
0.515	56	19.8	L	0.445	47.0	20.6	L
0.890	56	19.1	N	0.510	46	19.2	N

The reference diagrams as follows: (peak/average)





Fujian Inspection and Research Institute for Product Quality TEST REPORT

PAGE 12 OF 32

NO. (2020)DZI0022

4 Immunity

Standard compliance/severity level/criteria

Clause	Tests standard No.	Test specification level/Test mode	Test ports	Criteria
Table 1 1.1	Power frequency magnetic field immunity EN 61000-4-8	50Hz or 60Hz, 3A/m(r.m.s) μ s	Enclosure	A
Table 1 1.2	Radio-frequency electromagnetic field EN 61000-4-3	80MHz to 1000MHz 3V/m(unmodulated, r.m.s), 1kHz,80% AM modulated	Enclosure	A
Table 1 1.3		1,4GHz to 2,0GMHz 3V/m(unmodulated, r.m.s), 1kHz,80% AM modulated	Enclosure	A
Table 1 1.4		2,0GHz to 2,7GMHz 1V/m(unmodulated, r.m.s), 1kHz,80% AM modulated	Enclosure	A
Table 1 1.5	Electrostatic discharge EN 61000-4-2	± 8 kV air discharge ± 4 kV contact discharge (Direct Mode)	Enclosure	B
		± 4 kV HCP discharge ± 4 kV VCP discharge (Indirect Mode)	Enclosure	B
Table 2 2.1	Radio-frequency common mode EN 61000-4-6	0.15MHz to 80MHz 3V(unmodulated, r.m.s), 1kHz 80%, AM 150ohm source impedance	Signal ports (NOTE 1) (NOTE 2)	A
Table 3 3.1			Input and output DC power ports (NOTE 1) (NOTE 2)	A
Table 4 4.1			Input and output AC power ports (NOTE 1)	A
Table 2 2.2	Electrical fast transient/burst immunity EN 61000-4-4	± 0.5 kV(peak) 5/50ns Tr/Th 5kHz Repetition Frequency	Signal ports (NOTE 2)	B
Table 3 3.3		± 0.5 kV(peak) 5/50ns Tr/Th 5kHz Repetition Frequency	Input and output DC power ports (NOTE 3)	B
Table 4 4.5		± 1 kV(peak) 5/50ns Tr/Th 5kHz Repetition Frequency	Input and output AC power ports	B



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 13 OF 32

NO. (2020)DZI0022

Clause	Tests standard No.	Test specification level/Test mode	Test ports	Criteria
Table 3 3.2	Surges EN 61000-4-5	Line to reference ground for each individual line: ± 0.5 kV 1.2/50(8/20) Tr/Th μ s	Input and output DC power ports (NOTE 4)	B
Table 4 4.4		± 1 kV (peak) 1.2/50(8/20) Tr/Th μ s (line to line) ± 2 kV (peak) 1.2/50(8/20) Tr/Th μ s (line to earth or ground)	Input and output AC power ports	B
Table 4 4.2 4.3	Voltage dips EN 61000-4-11	Residual voltage 0% 0.5 period Residual voltage 0% 1 periods Residual voltage 70% 25/30 periods at 50/60Hz	AC power ports (NOTE 5)	B B C
	Voltage interruptions EN 61000-4-11	Residual voltage 0% 250/300 periods at 50/60Hz	AC power ports (NOTE 5)	C

NOTE:

- 1) The test level can also be defined as the equivalent current into a 150Ω load. Applicable only to ports which, according to the manufacturer's specification, support cable lengths greater than 3m.
- 2) Applicable only to ports which, according to the manufacturer's specification, may connect directly to outdoor cables.
- 3) Not applicable to input ports intended for connection to a battery or a rechargeable battery which must be removed or disconnected from the apparatus for recharging. Apparatus with a DC power input port intended for use with an AC-DC power adaptor shall be tested on the AC power input of the AC-DC power adaptor specified by the manufacturer or, where none is so specified, using a typical AC-DC power adaptor. The test is applicable to DC power input ports intended to be connected permanently to cables longer than 3m.
- 4) Not applicable to input ports intended for connection to a battery or a rechargeable battery which must be removed or disconnected from the apparatus for recharging. Apparatus with a DC power input port intended for use with an AC-DC power adaptor shall be tested on the AC power input of the AC-DC power adaptor specified by the manufacturer or, where none is so specified, using a typical AC-DC power adaptor. DC ports which are not intended to be connected to a DC distribution network are treated as signal ports.
- 5) Applicable only to input ports.



**Fujian Inspection and Research Institute
for Product Quality
TEST REPORT**

PAGE 14 OF 32

NO. (2020)DZI0022

Performance criterion

Criterion A	The apparatus shall continue to operate as intended during and after the test. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer, when the apparatus is used as intended. The performance level may be replaced by a permissible loss of performance. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, either of these may be derived from the product description and documentation and what the user may reasonably expect from the apparatus if used as intended.
Criterion B	The apparatus shall continue to operate as intended after the test. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer, when the apparatus is used as intended. The performance level may be replaced by a permissible loss of performance. During the test, degradation of performance is however allowed. No change of actual operating state or stored data is allowed. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, either of these may be derived from the product description and documentation and what the user may reasonably expect from the apparatus if used as intended.
Criterion C	Temporary loss of function is allowed, provided the function is self-recoverable or can be restored by the operation of the controls.



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 15 OF 32

NO. (2020)DZI0022

4.3 Electrostatic discharges

Standard: EN 61000-6-1
Test result: Pass
Severity: $\pm 4\text{kV}$ for Direct & Indirect Contact Discharge
 $\pm 8\text{kV}$ for Air Discharge

Performance criterion: B

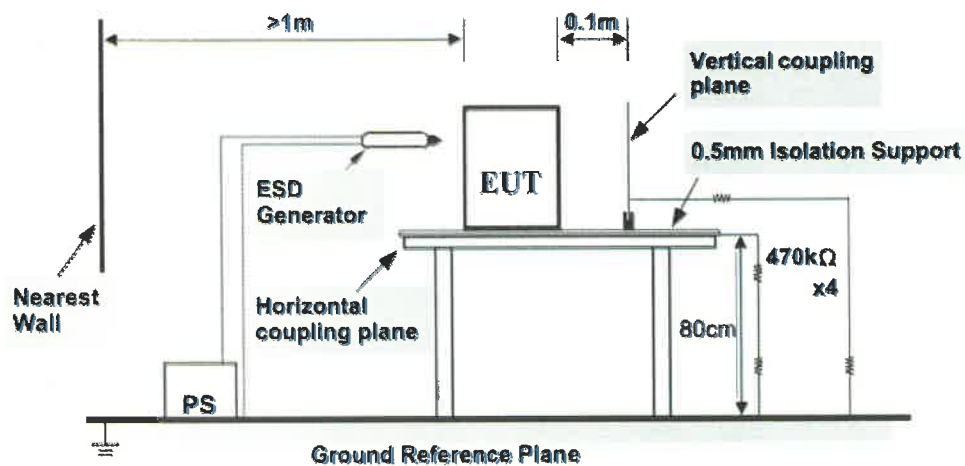
Temperature: $(17\sim 23)^\circ\text{C}$
Relative humidity: $(50\sim 60)\%$
Test date: 2020-05-26

Test voltage: AC 220V, 50Hz
Operating mode: Mode 1

4.1.1. Test method

The test was performed in accordance with EN 61000-4-2.

4.1.2. Test setup





Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 16 OF 32

NO. (2020)DZI0022

4.1.3. Test Result

Model Name	GBL-6110AD	Test Voltage	AC 220V, 50Hz
Location	Discharge method	Test voltage kV	Individual results
HCP (Horizontal coupling plane)	Indirect contact	± 4	B
VCP (Vertical coupling plane)	Indirect contact	± 4	B
Metal part & Screws	Direct contact	± 4	B
Gap	Air	± 8	B
insulating surfaces	Air	± 8	B

Note:1).N/A-denotes test is not applicable in device.



Fujian Inspection and Research Institute for Product Quality TEST REPORT

PAGE 17 OF 32

NO. (2020)DZ10022

4.2 Radio-frequency electromagnetic field

Standard: EN 61000-6-1
Test result: Pass
Severity: 3V/m for 80 MHz to 1000 MHz
3V/m for 1,4 GHz to 2,0 GHz
1V/m for 2,0 GHz to 2,7 GHz

Performance criterion: A

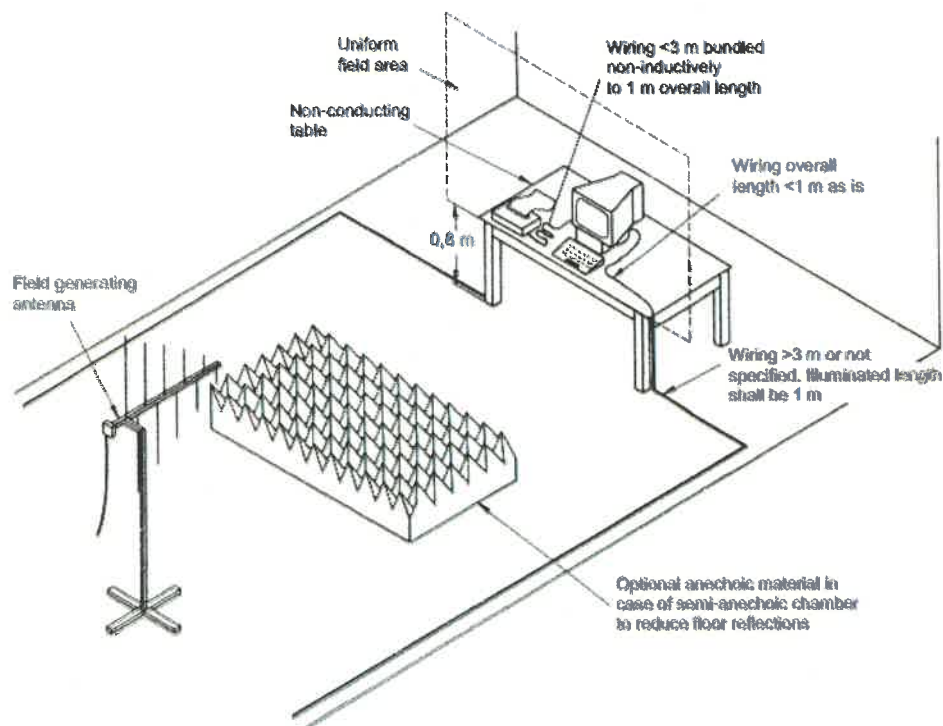
Temperature: (17~23) °C
Relative humidity: (50~60)%
Test date: 2020-05-26

Test voltage: AC 220V, 50Hz
Operating mode: Mode 1

4.2.1. Test method

The test was performed in accordance with EN 61000-4-3.

4.2.1 Test setup





Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 18 OF 32

NO. (2020)DZI0022

4.2.1 Test Result

Model Name		GBL-6110AD		Test Voltage		AC 220V, 50Hz	
Frequency MHz	Field strength V/m	Dwell time s	Frequency step	Polarity	Face	Individual results	
80~1000	3	3	1%	Horizontal	0°	A	
					90°	A	
					180°	A	
					270°	A	
				Vertical	0°	A	
					90°	A	
					180°	A	
					270°	A	
1400~2000	3	3	1%	Horizontal	0°	A	
					90°	A	
					180°	A	
					270°	A	
				Vertical	0°	A	
					90°	A	
					180°	A	
					270°	A	
2000~2700	1	3	1%	Horizontal	0°	A	
					90°	A	
					180°	A	
					270°	A	
				Vertical	0°	A	
					90°	A	
					180°	A	
					270°	A	

Note:1).N/A-denotes test is not applicable in device.



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 19 OF 32

NO. (2020)DZI0022

4.3 Radio-frequency common mode

Standard: EN 61000-6-1
Test result: Pass
Severity: 0.15MHz to 80MHz-3V(unmodulated, r.m.s),
1kHz 80%, AM

Performance criterion: A

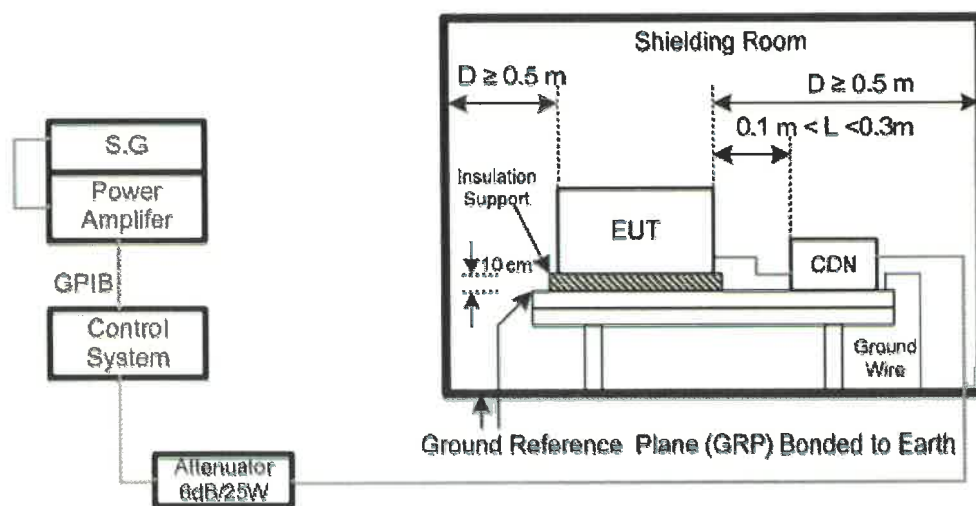
Temperature: (17~23) °C
Relative humidity: (50~60) %
Test date: 2020-05-28

Test voltage: AC 220V, 50Hz
Operating mode: Mode 1

4.3.1 Test method

The test was performed in accordance with EN 61000-4-6.

4.3.2 Test setup





Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 20 OF 32

NO. (2020)DZI0022

4.3.3 Test Result

Model Name	GBL-6110AD		Test Voltage	AC 220V, 50Hz	
Port	Frequency MHz	Voltage V	Dwell time s	Frequency step	Individual results
AC mains power ports	0.15~80	3	3	1%	A
DC network power ports	0.15~80	3			N/A
Analogue/digital data ports	0.15~80	3			N/A

Note:1).N/A-denotes test is not applicable in device.



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 21 OF 32

NO. (2020)DZI0022

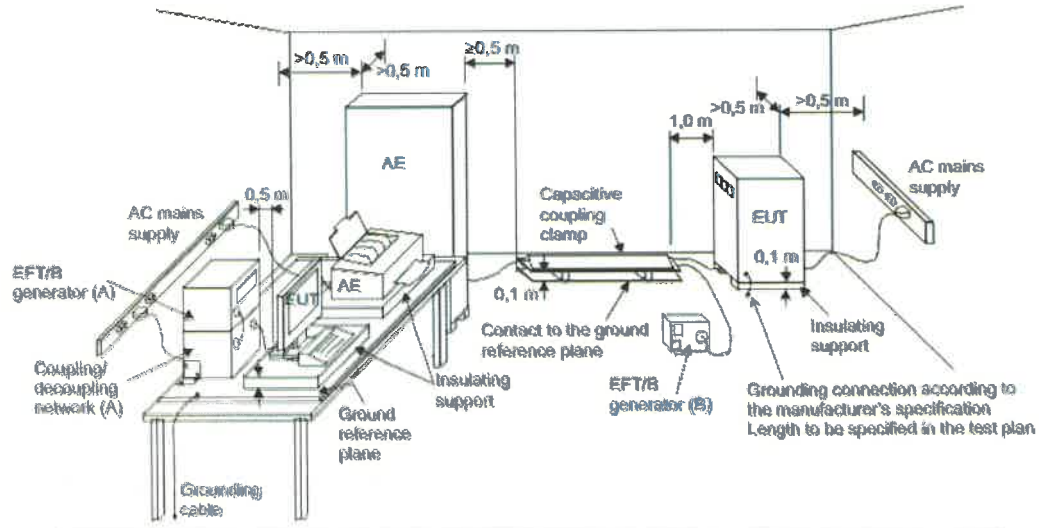
4.4 Electrical fast transient

Standard:	EN 61000-6-1
Test result:	Pass
Severity:	AC mains power ports : $\pm 1\text{kV}$ Analogue/digital data ports : $\pm 0.5\text{kV}$ DC network power ports : $\pm 0.5\text{kV}$
Performance criterion:	B
Temperature:	(17~23) $^{\circ}\text{C}$
Relative humidity:	(50~60)%
Test date:	2020-05-28
Test voltage:	AC 220V, 50Hz
Operating mode	Mode 1

4.4.1 Test method

The test was performed in accordance with EN 61000-4-4.

4.4.2 Test setup





Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 22 OF 32

NO. (2020)DZI0022

4.4.3 Test Result

Model Name	GBL-6110AD	Test Voltage	AC 220V, 50Hz
------------	------------	--------------	---------------

Conductor		Voltage peak kV	Repetition rate kHz	Individual results
AC mains power ports	Live -Neutral - Earth	± 1	5	B
DC network power ports	Positive - negative	± 0.5		N/A
Analogue/digital data ports	Clamp	± 0.5		N/A

Note:1).N/A-denotes test is not applicable in device.



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 23 OF 32

NO. (2020)DZI0022

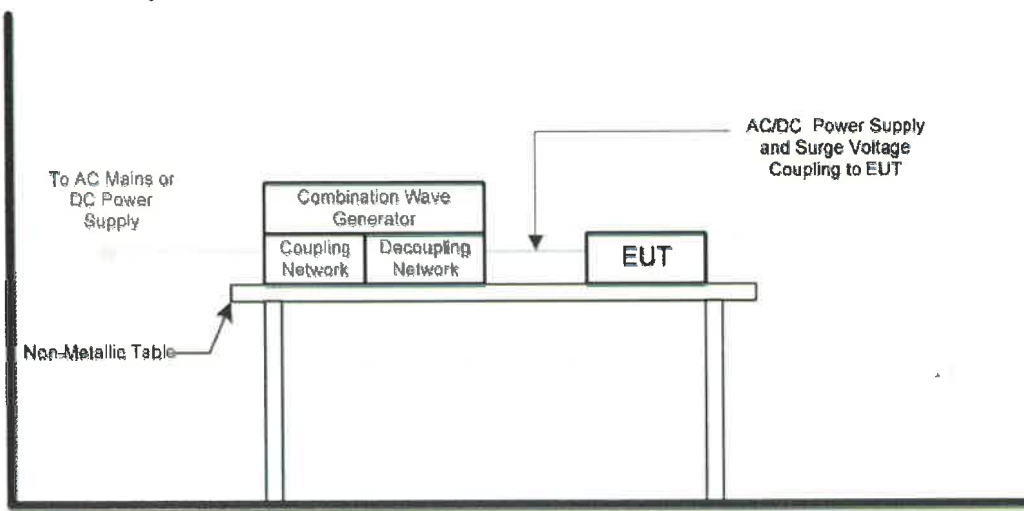
4.5 Surges

Standard:	EN 61000-6-1
Test result:	Pass
Severity:	AC mains power ports: $\pm 1\text{kV}$ (between live & neutral) $\pm 2\text{kV}$ (between live & earth) $\pm 2\text{kV}$ (between neutral & earth) DC network power ports: $\pm 0.5\text{kV}$ (between Positive & negative)
Performance criterion:	B
Temperature:	(17~23) °C
Relative humidity:	(50~60) %
Test date:	2020-05-26
Test voltage:	AC 220V, 50Hz
Operating mode:	Mode 1

4.5.1 Test method

The test was performed in accordance with EN 61000-4-5.

4.5.2 Test setup





Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 24 OF 32

NO. (2020)DZI0022

4.5.3 Test Result

Model Name		GBL-6110AD		Test Voltage		AC 220V, 50Hz	
Conductor		Voltage kV	No. of surge	Tr/Th μ s	Surge interval s	Phase angle	Individual results
AC mains power ports	Live - Neutral	± 1	5	1.2/50	60	90,270	B
	Live -Earth	± 2	5	1.2/50	60	90,270	N/A
	Neutral -Earth	± 2	5	1.2/50	60	90,270	N/A
DC network power ports	Positive -negative	± 0.5	5	1.2/50	60	/	N/A

Note:1).N/A-denotes test is not applicable in device.



Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 25 OF 32

NO. (2020)DZI0022

4.6 Voltage dips and interruptions

Standard: EN 61000-6-1
Test result: Pass
Severity: 0.5, 1 and 250 periods for 0% of U_T
25 periods for 70% of U_T

Performance criterion: B for 0% of U_T for 0.5 period
B for 0% of U_T for 1 period
C for 70% of U_T for 25 period
C for 0% of U_T for 250 period

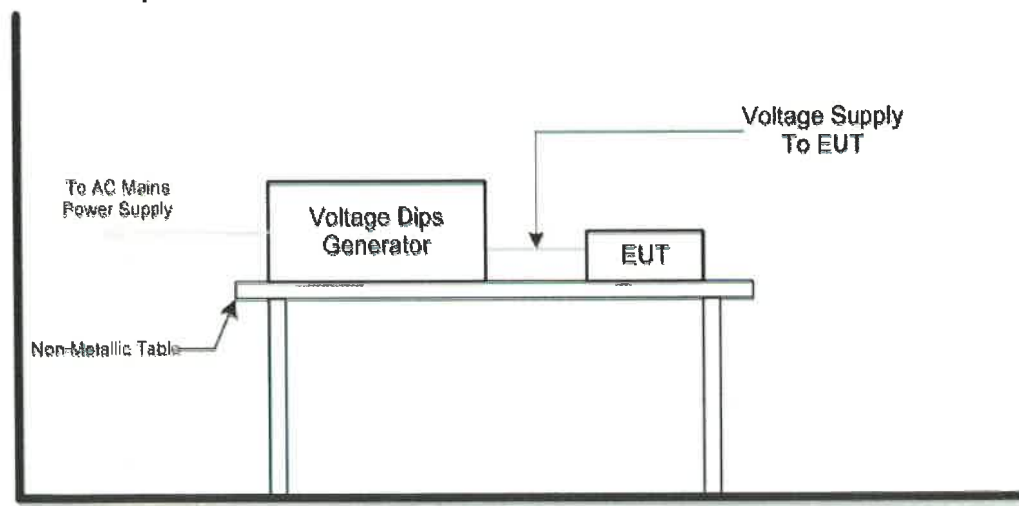
Temperature: (17~23) °C
Relative humidity: (50~60) %
Test date: 2020-05-28

Test voltage: AC 220V, 50Hz

4.6.1 Test method

The test was performed in accordance with EN 61000-4-11.

4.6.2 Test setup





Fujian Inspection and Research Institute for Product Quality

TEST REPORT

PAGE 26 OF 32

NO. (2020)DZI0022

4.6.3 Test Result

Model Name	GBL-6110AD	Test Voltage	AC 220V, 50Hz	
AC Input	Test level (% of U_T)	Duration (period)	Angel of phase	Individual results
AC220V,50Hz	0	0.5	0°	B
	0	1	0°	B
	70	25	0°	B
	0	250	0°	C

Note:1).N/A-denotes test is not applicable in device.



**Fujian Inspection and Research Institute
for Product Quality
TEST REPORT**

PAGE 27 OF 32

NO. (2020)DZ10022

Appendix A Equipments used during test

DESCRIPTION	MANUFACTURER	MODEL NO./TYPE	Management NO.	NEXT CAL
Shielded Room	ETS Lindgren	8.8m×4.8m×3m	DZ-27-006-7-1	2023-03-19
Semi-Anechoic Chamber	Frankonia	22.28m×15.98m×9.6m	DZ-25-108-1	2025-03-23
EMI Test Receiver	R&S	ESCI	DZ-25-082	2021-03-16
Artificial-Mains Network	SCHAFFNER	NNB42	DZ-25-075	2021-09-17
EMI Test Receiver	R&S	ESU	DZ-25-105-1	2021-03-16
Biconilog Antenna	ETS-LINDGREN	3142D	DZ-25-115	2021-06-28
Pre-Amplifier	R/S	SCU-01	DZ-25-105-3	2021-04-11
Power Analyzer/AC Power Source	California Instruments	PACS-1/5001ix	DZ-25-067	2021-04-10
ESD Simulator	EMTEST	NX30	DZ-25-203	2020-10-30
Fully anechoic chamber	ETS Lindgren	8m×4m×3.8m	DZ-25-043-2	2021-03-01
RS Test System	R&S	TS9981	DZ-25-134	2021-03-10
Turntable	Sunol Sciences	FM1511	DZ-25-048	N/A
Turntable/ Antenna Positioning Tower	Sunol Sciences	SC99V	DZ-25-040	N/A
Electric fast transient pulse generator	EMTEST	EFT500N8.1	DZ-25-130	2020-09-08
CS Test System	TESEQ	NSG4070	DZ-25-117	2021-03-05
Amplifier	AR	100A250A	DZ-01-090	N/A
Power-line coupled decoupling network	FCC	4412-25(two line of 25A)	DZ-25-034	2021-05-21
Injection clamp	FCC	EM-101	DZ-25-026	N/A
EMC immunity Tester	Keytek	EMC Pro	DZ-25-015	2021-03-08
NOTE N/A - not applicable				

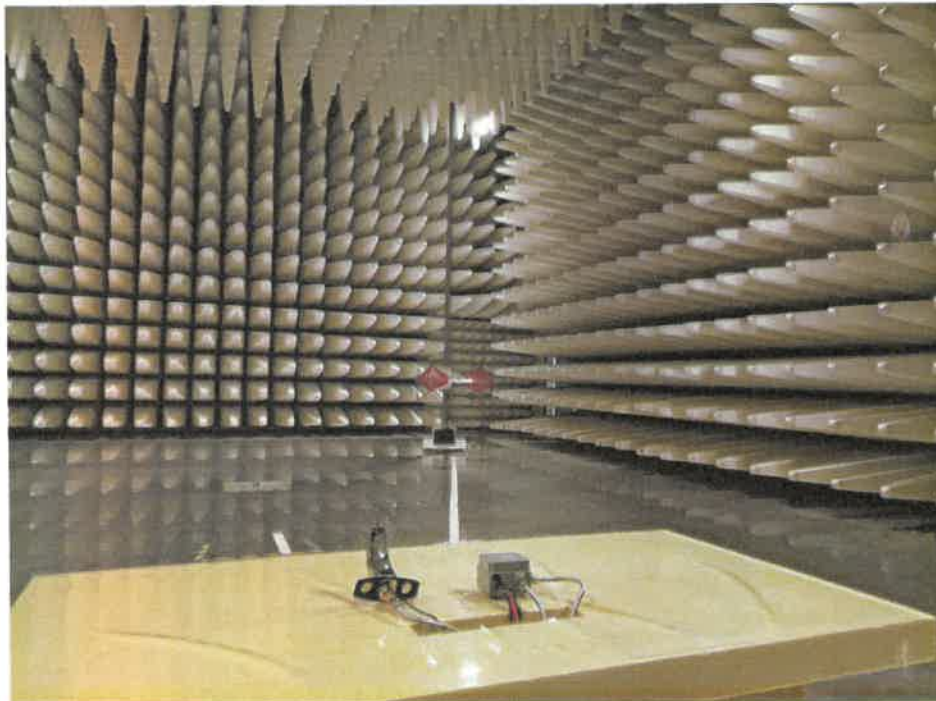


**Fujian Inspection and Research Institute
for Product Quality
TEST REPORT**

PAGE 28 OF 32

NO. (2020)DZI0022

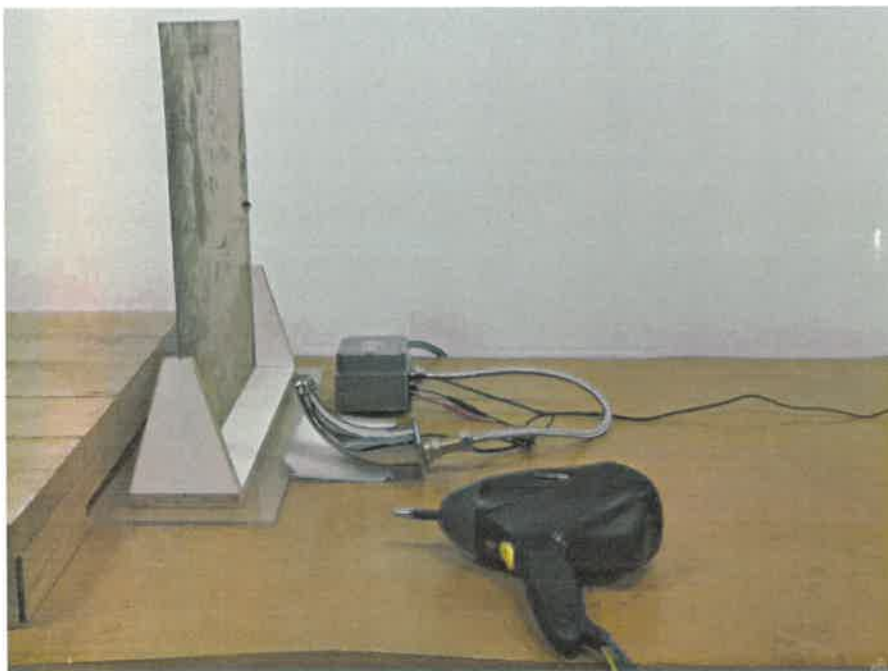
Appendix B EUT test photo



Radiated emissions on enclosure port



Conducted emissions on low voltage AC mains port



Electrostatic discharges



Radio-frequency electromagnetic field



**Fujian Inspection and Research Institute⁴
for Product Quality
TEST REPORT**

PAGE 30 OF 32

NO. (2020)DZI0022



Radio-frequency common mode



Electrical fast transient



**Fujian Inspection and Research Institute
for Product Quality
TEST REPORT**

PAGE 31 OF 32

NO. (2020)DZI0022



Surges



Voltage dips and interruptions



**Fujian Inspection and Research Institute
for Product Quality
TEST REPORT**

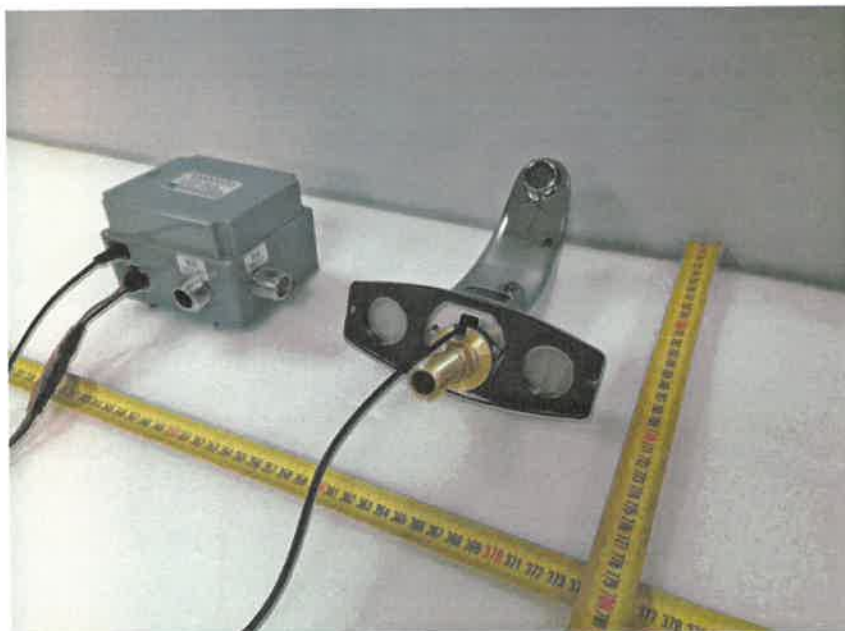
PAGE 32 OF 32

NO. (2020)DZI0022

Appendix C Photo of EUT



GBL-6110AD



**GBL-6110AD
--End of report--**

(3) 合格
合格